VCF/400 クロス検証テスト結果レポート

項目	内容
対象	オリジナル RPG (PUB400 `ASHIBATA2`) / Java Web 版 (`deliverables/java/`, PR #11) / iPad 版 (`deliverables/ipad/`, PR #13)
テストケース	`deliverables/tests/cases.json` (37 ケース)
実行方式	PUB400: tn5250 (`run_pub400.py`) / Java: Playwright Chromium (`run_java.js`) / iPad: XCUITest test-without-building (`run_ipad.sh`, 録画は `record_ipad.sh`)
判定	`expect` の全キーが 3 環境の `observed` と一致 = PASS

1. サマリ

環境	PASS	FAIL / 未実行
PUB400	37 / 37	0
Java Web	37 / 37	0
iPad	37 / 37	0
3 環境一致	**37 / 37**	0

1.1 分類別

分類	内容	ケース数	3 環境一致
ADDVOTE	投票	13	13
ADDGBCMT	ゲストブック記入	6	6
READGBCMT	ゲストブック閲覧	7	7
LRN400	LEARN/400	4	4
EXHBMENU	展示キオスク	7	7

2. 不一致と対処

本 run は、PUB400 を `ASHIBATA2` にソースビルドした RPG、Java を PR #11 (`devin/1791970110-vcf400-java-web`, `03ed528`)、iPad を PR #13 (`devin/1791970125-ipad-deliverables`, `e67316a`) として実行した。`compare.py` の最終結果は PUB400 37/37、Java 37/37、iPad 37/37、3 環境一致 37/37 で、不一致は 0 件である。

Java の最終コミット `32ead03` は VCFMAIN 見出し行の空白保持 (CSS `white-space: pre`) のみの変更で、37 ケースの判定対象には影響しない (Maven 53/53 は `03ed528` 以降も通過)。

実行前後の初期化: PUB400 は `pub400_db.py reset` (`logs/pub400_seed.sql` の全 6 表) を開始前と終了後に必ず実行、Java は `VCF_DB_API=true` の fresh H2 + DB API reset、iPad は `-VCF_RESET` 起動引数でケースごとに SQLite を初期化した。アプリケーションと runner に、一致させるための個別の補正は入れていない。

ブラウザ E2E (testing agent) で、Java の VCFMAIN に台帳 L-01-01 の見出し `WELCOME TO...` / `V I N T A G E C O M P U T E R F E S T I V A L 2 0 2 4` が欠けていることが見つかり、`03ed528` で追加した (cases.json の判定外の表示要素)。iPad は当初から表示していた。

最終実行における不一致は 0 件。

UI 日本語化とクロス検証契約

比較で参照する `#errline` / `errline` 識別子は英語原文 M-xx を返し、画面上の日本語は `.errline-ja` / `errline_ja` に表示する。画面検出のアンカーとして、台帳で保持対象とした legacy title の英語文字列は残す。

3. ケース一覧

入力、期待値、3 環境の実測値、および合否を以下に示す。

ID	名称	入力	期待値	PUB400 実測	Java 実測	iPad 実測	合否
CV-01	全項目未入力 (展示 ID は保護・事前入力)	{"badge":"","award":""}	{"screen":"VOTE1","errline":"Must enter Award ID","vote":null}	{"screen":"VOTE1","errline":"Must enter Award ID","vote":null}	{"screen":"VOTE1","errline":"Must enter Award ID","vote":null}	{"errline":"Must enter Award ID","screen":"VOTE1","vote":null}	PASS
CV-02	アワード未入力	{"badge":"7001","award":""}	{"screen":"VOTE1","errline":"Must enter Award ID","vote":null}	{"screen":"VOTE1","errline":"Must enter Award ID","vote":null}	{"screen":"VOTE1","errline":"Must enter Award ID","vote":null}	{"errline":"Must enter Award ID","screen":"VOTE1","vote":null}	PASS
CV-03	バッジ未入力	{"badge":"","award":"1"}	{"screen":"VOTE1","errline":"Must enter badge number","vote":null}	{"screen":"VOTE1","errline":"Must enter badge number","vote":null}	{"screen":"VOTE1","errline":"Must enter badge number","vote":null}	{"errline":"Must enter badge number","screen":"VOTE1","vote":null}	PASS
CV-04	正常投票 (新規バッジ 7001, アワード 1)	{"badge":"7001","award":"1"}	{"screen":"VOTEEND","vote":{"badge":7001,"award":1,"exhibit":"ASHIBATA"}}	{"screen":"VOTEEND","vote":{"badge":7001,"award":1,"exhibit":"ASHIBATA"}}	{"screen":"VOTEEND","vote":{"badge":7001,"award":1,"exhibit":"ASHIBATA"}}	{"screen":"VOTEEND","vote":{"award":1,"badge":7001,"exhibit":"ASHIBATA"}}	PASS
CV-05	重複バッジ (7001 で再投票)	{"badge":"7001","award":"2"}	{"screen":"VOTE1","errline":"You have already voted.","vote":{"badge":7001,"award":1,"exhibit":"ASHIBATA"}}	{"screen":"VOTE1","errline":"You have already voted.","vote":{"badge":7001,"award":1,"exhibit":"ASHIBATA"}}	{"screen":"VOTE1","errline":"You have already voted.","vote":{"badge":7001,"award":1,"exhibit":"ASHIBATA"}}	{"errline":"You have already voted.","screen":"VOTE1","vote":{"award":1,"badge":7001,"exhibit":"ASHIBATA"}}	PASS
CV-06	存在しないアワード (9)	{"badge":"7002","award":"9"}	{"screen":"VOTE1","errline":"Award does not exist.","vote":null}	{"screen":"VOTE1","errline":"Award does not exist.","vote":null}	{"screen":"VOTE1","errline":"Award does not exist.","vote":null}	{"errline":"Award does not exist.","screen":"VOTE1","vote":null}	PASS
CV-07	境界値: アワード 0 (数値 0 = 未入力扱い)	{"badge":"7002","award":"0"}	{"screen":"VOTE1","errline":"Must enter Award ID","vote":null}	{"screen":"VOTE1","errline":"Must enter Award ID","vote":null}	{"screen":"VOTE1","errline":"Must enter Award ID","vote":null}	{"errline":"Must enter Award ID","screen":"VOTE1","vote":null}	PASS
CV-08	共用端末: 展示 ID 未入力	{"badge":"7002","exhibit":"","award":"1"}	{"screen":"VOTE1","errline":"Must enter Exhibit ID","vote":null}	{"screen":"VOTE1","errline":"Must enter Exhibit ID","vote":null}	{"screen":"VOTE1","errline":"Must enter Exhibit ID","vote":null}	{"errline":"Must enter Exhibit ID","screen":"VOTE1","vote":null}	PASS
CV-09	共用端末: 存在しない展示 NOSUCH	{"badge":"7002","exhibit":"NOSUCH","award":"1"}	{"screen":"VOTE1","errline":"Exhibit does not exist","vote":null}	{"screen":"VOTE1","errline":"Exhibit does not exist","vote":null}	{"screen":"VOTE1","errline":"Exhibit does not exist","vote":null}	{"errline":"Exhibit does not exist","screen":"VOTE1","vote":null}	PASS

ID	名称	入力	期待値	PUB400 実測	Java 実測	iPad 実測	合否
CV-10	共用端末: 資格外展示 NOVOTE (ELIGIBLE=0)	{"badge":"7002","exhibit":"NOVOTE","award":"1"}	{"screen":"VOTE1","errline":"Exhibit ineligible for award","vote":null}	{"screen":"VOTE1","errline":"Exhibit ineligible for award","vote":null}	{"screen":"VOTE1","errline":"Exhibit ineligible for award","vote":null}	{"errline":"Exhibit ineligible for award","screen":"VOTE1","vote":null}	PASS
CV-11	共用端末: 正常投票 (7002, DEMO400, アワード 2)	{"badge":"7002","exhibit":"DEMO400","award":"2"}	{"screen":"VOTEEND","vote":{"badge":7002,"award":2,"exhibit":"DEMO400"}}	{"screen":"VOTEEND","vote":{"badge":7002,"award":2,"exhibit":"DEMO400"}}	{"screen":"VOTEEND","vote":{"badge":7002,"award":2,"exhibit":"DEMO400"}}	{"screen":"VOTEEND","vote":{"award":2,"badge":7002,"exhibit":"DEMO400"}}	PASS
CV-12	複合エラー (重複バッジ+存在しない展示+存在しないアワード) → 最後の検査が勝つ	{"badge":"7001","exhibit":"NOSUCH","award":"9"}	{"screen":"VOTE1","errline":"Award does not exist.","vote":{"badge":7001,"award":1,"exhibit":"ASHIBATA"}}	{"screen":"VOTE1","errline":"Award does not exist.","vote":{"badge":7001,"award":1,"exhibit":"ASHIBATA"}}	{"screen":"VOTE1","errline":"Award does not exist.","vote":{"badge":7001,"award":1,"exhibit":"ASHIBATA"}}	{"errline":"Award does not exist.","screen":"VOTE1","vote":{"award":1,"badge":7001,"exhibit":"ASHIBATA"}}	PASS
CV-13	投票停止 ALWVOTE='N' → ENDOFCON	{"badge":"7003","award":"1"}	{"screen":"ENDOFCON","vote":null}	{"screen":"ENDOFCON","vote":null}	{"screen":"ENDOFCON","vote":null}	{"screen":"ENDOFCON","vote":null}	PASS
CV-14	全項目未入力 (展示 ID は保護・事前入力)	{"name":"","comment":""}	{"screen":"ADDCMT","errline":"Must enter a comment","comment":{"id":3,"row":null}}	{"screen":"ADDCMT","errline":"Must enter a comment","comment":{"id":3,"row":null}}	{"screen":"ADDCMT","errline":"Must enter a comment","comment":{"id":3,"row":null}}	{"comment":{"id":3,"row":null},"errline":"Must enter a comment","screen":"ADDCMT"}	PASS
CV-15	コメント未入力	{"name":"Devin Tester","comment":""}	{"screen":"ADDCMT","errline":"Must enter a comment","comment":{"id":3,"row":null}}	{"screen":"ADDCMT","errline":"Must enter a comment","comment":{"id":3,"row":null}}	{"screen":"ADDCMT","errline":"Must enter a comment","comment":{"id":3,"row":null}}	{"comment":{"id":3,"row":null},"errline":"Must enter a comment","screen":"ADDCMT"}	PASS
CV-16	名前未入力	{"name":"","comment":"Hello"}	{"screen":"ADDCMT","errline":"Must enter your name","comment":{"id":3,"row":null}}	{"screen":"ADDCMT","errline":"Must enter your name","comment":{"id":3,"row":null}}	{"screen":"ADDCMT","errline":"Must enter your name","comment":{"id":3,"row":null}}	{"comment":{"id":3,"row":null},"errline":"Must enter your name","screen":"ADDCMT"}	PASS
CV-17	正常記入 → ENDCMT, CMTID=3 (最終+1), VISIBLE=Y	{"name":"Devin Tester","comment":"Cross validation run"}	{"screen":"ENDCMT","comment":{"id":3,"row":{"id":3,"visible":"Y","exhibit":"ASHIBATA","name":"Devin Tester","comment":"Cross validation run"}}}	{"screen":"ENDCMT","comment":{"id":3,"row":{"id":3,"visible":"Y","exhibit":"ASHIBATA","name":"Devin Tester","comment":"Cross validation run"}}}	{"screen":"ENDCMT","shownId":3,"comment":{"id":3,"row":{"id":3,"visible":"Y","exhibit":"ASHIBATA","name":"Devin Tester","comment":"Cross validation run"}}}	{"comment":{"id":3,"row":{"comment":"Cross validation run","exhibit":"ASHIBATA","id":3,"name":"Devin Tester","visible":"Y"}},"screen":"ENDCMT","shownId":3}	PASS
CV-18	共用端末: 展示 ID 未入力	{"name":"Guest","exhibit":"","comment":"x"}	{"screen":"ADDCMT","errline":"Must enter Exhibit ID","comment":{"id":4,"row":null}}	{"screen":"ADDCMT","errline":"Must enter Exhibit ID","comment":{"id":4,"row":null}}	{"screen":"ADDCMT","errline":"Must enter Exhibit ID","comment":{"id":4,"row":null}}	{"comment":{"id":4,"row":null},"errline":"Must enter Exhibit ID","screen":"ADDCMT"}	PASS

ID	名称	入力	期待値	PUB400 実測	Java 実測	iPad 実測	合否
CV-19	共用端末: 存在しない展示 NOSUCH でも記入できる (存在チェックなし)	{"name":"Guest","exhibit":"NOSUCH","comment":"Unchecked exhibit"}	{"screen":"ENDCMT","comment":{"id":4,"row":{"id":4,"visible":"Y","exhibit":"NOSUCH","name":"Guest","comment":"Unchecked exhibit"}}}	{"screen":"ENDCMT","comment":{"id":4,"row":{"id":4,"visible":"Y","exhibit":"NOSUCH","name":"Guest","comment":"Unchecked exhibit"}}}	{"screen":"ENDCMT","shownId":4,"comment":{"id":4,"row":{"id":4,"visible":"Y","exhibit":"NOSUCH","name":"Guest","comment":"Unchecked exhibit"}}}	{"comment":{"id":4,"row":{"comment":"Unchecked exhibit","exhibit":"NOSUCH","id":4,"name":"Guest","visible":"Y"}},"screen":"ENDCMT","shownId":4}	PASS
CV-20	コメント ID 未入力	{"cmtid":""}	{"screen":"READCMT","errline":"Must enter CommentID"}	{"screen":"READCMT","total":4,"errline":"Must enter CommentID","out":{"name":"","title":"","cmt":""}}	{"screen":"READCMT","total":4,"errline":"Must enter CommentID"}	{"errline":"Must enter CommentID","screen":"READCMT","total":4}	PASS
CV-21	ID 1 → 名前 11 桁 / 展示タイトル / コメント	{"cmtid":"1"}	{"screen":"READCMT","total":4,"out":{"name":"Great exhib","title":"IBM i on PUB400 Demo","cmt":"VCF/400 running on PUB400."}}	{"screen":"READCMT","total":4,"errline":"ASHIBATA","out":{"name":"Great exhib","title":"IBM i on PUB400 Demo","cmt":"VCF/400 running on PUB400."}}	{"screen":"READCMT","total":4,"errline":"","out":{"name":"Great exhib","title":"IBM i on PUB400 Demo","cmt":"VCF/400 running on PUB400."}}	{"errline":"","out":{"cmt":"VCF/400 running on PUB400.","name":"Great exhib","title":"IBM i on PUB400 Demo"},"screen":"READCMT","total":4}	PASS
CV-22	ID 3 (この実行で追加したコメント)	{"cmtid":"3"}	{"screen":"READCMT","total":4,"out":{"name":"Devin Teste","title":"IBM i on PUB400 Demo","cmt":"Cross validation run"}}	{"screen":"READCMT","total":4,"errline":"ASHIBATA","out":{"name":"Devin Teste","title":"IBM i on PUB400 Demo","cmt":"Cross validation run"}}	{"screen":"READCMT","total":4,"errline":"","out":{"name":"Devin Teste","title":"IBM i on PUB400 Demo","cmt":"Cross validation run"}}	{"errline":"","out":{"cmt":"Cross validation run","name":"Devin Teste","title":"IBM i on PUB400 Demo"},"screen":"READCMT","total":4}	PASS
CV-23	存在しない ID 9999 → 最終レコード (4, 他展示) → 帰属外メッセージ	{"cmtid":"9999"}	{"screen":"READCMT","total":4,"out":{"name":"","title":"","cmt":"This comment is not part of this guestbook."}}	{"screen":"READCMT","total":4,"errline":"ASHIBATA","out":{"name":"","title":"","cmt":"This comment is not part of this guestbook."}}	{"screen":"READCMT","total":4,"errline":"","out":{"name":"","title":"","cmt":"This comment is not part of this guestbook."}}	{"errline":"","out":{"cmt":"This comment is not part of this guestbook.","name":"","title":""},"screen":"READCMT","total":4}	PASS
CV-24	非表示コメント (ID 2 を VISIBLE=N)	{"cmtid":"2"}	{"screen":"READCMT","total":4,"out":{"name":"Name Hidden","title":"","cmt":"This comment hidden by an admin - offensive content."}}	{"screen":"READCMT","total":4,"errline":"ASHIBATA","out":{"name":"Name Hidden","title":"","cmt":"This comment hidden by an admin - offensive content."}}	{"screen":"READCMT","total":4,"errline":"","out":{"name":"Name Hidden","title":"","cmt":"This comment hidden by an admin - offensive content."}}	{"errline":"","out":{"cmt":"This comment hidden by an admin - offensive content.","name":"Name Hidden","title":""},"screen":"READCMT","total":4}	PASS
CV-25	他展示 (DEMO400) から ASHIBATA のコメント 1 を読む → 帰属外	{"cmtid":"1"}	{"screen":"READCMT","total":4,"out":{"name":"","title":"","cmt":"This comment is not part of this guestbook."}}	{"screen":"READCMT","total":4,"errline":"DEMO400","out":{"name":"","title":"","cmt":"This comment is not part of this guestbook."}}	{"screen":"READCMT","total":4,"errline":"","out":{"name":"","title":"","cmt":"This comment is not part of this guestbook."}}	{"errline":"","out":{"cmt":"This comment is not part of this guestbook.","name":"","title":""},"screen":"READCMT","total":4}	PASS

ID	名称	入力	期待値	PUB400 実測	Java 実測	iPad 実測	合否
CV-26	共用端末は全展示を閲覧可 / 展示タイトルは SETLL+READ (NOSUCH → NOVOTE)	{"cmtid":"4"}	{"screen":"READCMT","total":4,"out":{"name":"Guest","title":"Ineligible test exhibit (ELIGIBLE=0)","cmt":"Unchecked exhibit"}}	{"screen":"READCMT","total":4,"errline":"MM2024","out":{"name":"Guest","title":"Ineligible test exhibit (ELIGIBLE=0)","cmt":"Unchecked exhibit"}}	{"screen":"READCMT","total":4,"errline":"","out":{"name":"Guest","title":"Ineligible test exhibit (ELIGIBLE=0)","cmt":"Unchecked exhibit"}}	{"errline":"","out":{"cmt":"Unchecked exhibit","name":"Guest","title":"Ineligible test exhibit (ELIGIBLE=0)"},"screen":"READCMT","total":4}	PASS
CV-27	F5 → 2 ページ目	{"keys":["F5"]}	{"screen":"LRN400","page":2,"content":"The AS/400 was introduced by IBM in June 1988 ... page 2"}	{"screen":"LRN400","page":2,"content":"The AS/400 was introduced by IBM in June 1988 ... page 2"}	{"screen":"LRN400","page":2,"content":"The AS/400 was introduced by IBM in June 1988 ... page 2"}	{"content":"The AS/400 was introduced by IBM in June 1988 ... page 2","page":2,"screen":"LRN400"}	PASS
CV-28	F5, F8 → 1 ページ目に戻る	{"keys":["F5","F8"]}	{"screen":"LRN400","page":1,"content":"Welcome to LEARN/400! This is page 1. Press F5 for next page, F8 for previous, F3 to exit."}	{"screen":"LRN400","page":1,"content":"Welcome to LEARN/400! This is page 1. Press F5 for next page, F8 for previous, F3 to exit."}	{"screen":"LRN400","page":1,"content":"Welcome to LEARN/400! This is page 1. Press F5 for next page, F8 for previous, F3 to exit."}	{"content":"Welcome to LEARN/400! This is page 1. Press F5 for next page, F8 for previous, F3 to exit.","page":1,"screen":"LRN400"}	PASS
CV-29	F5, F5 → 3 ページ目 (EXTRA='END') で終了 → VCFMAIN	{"keys":["F5","F5"]}	{"screen":"VCFMAIN"}	{"screen":"VCFMAIN"}	{"screen":"VCFMAIN"}	{"screen":"VCFMAIN"}	PASS
CV-30	F3 → 終了 → VCFMAIN	{"keys":["F3"]}	{"screen":"VCFMAIN"}	{"screen":"VCFMAIN"}	{"screen":"VCFMAIN"}	{"screen":"VCFMAIN"}	PASS
CV-31	ASHIBATA (ELIGIBLE=1, ENLRN400=1) → 1,2,3,4	{}	{"screen":"KIOSK","options":["1","2","3","4"]}	{"screen":"KIOSK","options":["1","2","3","4"]}	{"screen":"KIOSK","options":["1","2","3","4"]}	{"options":["1","2","3","4"],"screen":"KIOSK"}	PASS
CV-32	NOVOTE (ELIGIBLE=0, ENLRN400=0) → 3,4 のみ	{}	{"screen":"KIOSK","options":["3","4"]}	{"screen":"KIOSK","options":["3","4"]}	{"screen":"KIOSK","options":["3","4"]}	{"options":["3","4"],"screen":"KIOSK"}	PASS
CV-33	DEMO400 (ELIGIBLE=1, ENLRN400=0) → 1,3,4	{}	{"screen":"KIOSK","options":["1","3","4"]}	{"screen":"KIOSK","options":["1","3","4"]}	{"screen":"KIOSK","options":["1","3","4"]}	{"options":["1","3","4"],"screen":"KIOSK"}	PASS
CV-34	隠しオプション 7 + 誤パスワード → キオスクメニューに戻る	{"option":"7","password":"WRONGPW"}	{"screen":"KIOSK","options":["1","2","3","4"],"path":["ADMPSWRD","KIOSK"]}	{"screen":"KIOSK","options":["1","2","3","4"],"path":["ADMPSWRD","KIOSK"]}	{"screen":"KIOSK","options":["1","2","3","4"],"path":["ADMPSWRD","KIOSK"]}	{"options":["1","2","3","4"],"path":["ADMPSWRD","KIOSK"],"screen":"KIOSK"}	PASS

ID	名称	入力	期待値	PUB400 実測	Java 実測	iPad 実測	合否
CV-35	隠しオプション 7 + 正パスワード (SETTINGS.ADMPSWRD) → キオスク終了 → VCFMAIN	{"option":"7","password":"VCF2024"}	{"screen":"VCFMAIN","options":["1","2","3","4"],"path":["ADMPSWRD","VCFMAIN"]}	{"screen":"VCFMAIN","options":["1","2","3","4"],"path":["ADMPSWRD","VCFMAIN"]}	{"screen":"VCFMAIN","options":["1","2","3","4"],"path":["ADMPSWRD","VCFMAIN"]}	{"options":["1","2","3","4"],"path":["ADMPSWRD","VCFMAIN"],"screen":"VCFMAIN"}	PASS
CV-36	オプション 1 → NTRSTIT → VOTE1 (展示 ASHIBATA 固定) → F12 でキオスクへ戻る	{"option":"1"}	{"screen":"KIOSK","options":["1","2","3","4"],"path":["NTRSTIT","VOTE1","KIOSK"],"exhibit":"ASHIBATA"}	{"screen":"KIOSK","options":["1","2","3","4"],"exhibit":"ASHIBATA","path":["NTRSTIT","VOTE1","KIOSK"]}	{"screen":"KIOSK","options":["1","2","3","4"],"exhibit":"ASHIBATA","path":["NTRSTIT","VOTE1","KIOSK"]}	{"exhibit":"ASHIBATA","options":["1","2","3","4"],"path":["NTRSTIT","VOTE1","KIOSK"],"screen":"KIOSK"}	PASS
CV-37	資格外展示でオプション 1 を入力 → 無視 (キオスクメニューのまま)	{"option":"1"}	{"screen":"KIOSK","options":["3","4"],"path":["KIOSK"]}	{"screen":"KIOSK","options":["3","4"],"path":["KIOSK"]}	{"screen":"KIOSK","options":["3","4"],"path":["KIOSK"]}	{"options":["3","4"],"path":["KIOSK"],"screen":"KIOSK"}	PASS

4. 実行環境

項目	値
PUB400 / Java 実行ホスト	Linux (Ubuntu 22.04)
iPad 実行ホスト	macOS + Xcode (iOS Simulator SDK 26.5), iPad Pro 13-inch (M5) シミュレータ
Java	OpenJDK 17.0.19 / Spring Boot / H2 (MODE=DB2)
tn5250	0.19.0
Playwright	1.58.2 (Chromium)
PUB400	`ASHIBATA1` / `ASHIBATA2`
実行日	2026-10-14

5. 動画

環境	動画パス	長さ	解像度	サイズ	収録内容	速度
PUB400	`deliverables/video/1_PUB400_RPG.mp4`	492.50 秒	1280x720 (h264)	8.59 MB	実 5250 画面フレーム連続表示	等速
Java Web	`deliverables/video/2_Java_Web.mp4`	150.40 秒	1280x720 (h264)	6.61 MB	Chromium 実画面	等速
iPad	`deliverables/video/3_iPad.mp4`	376.30 秒	1280x720 (h264)	6.78 MB	シミュレータ実画面	等速

3 本とも実時間の等速録画であり、追加の速度変更は行っていない。PUB400 は `self.s.screen.display` から取得した 5250 画面フレームを連続表示し、Java は Chromium の実画面、iPad はシミュレータの実画面を収録した。iPad のケース帯 オーバーレイは ffmpeg に `drawtext` がないため Playwright で PNG を生成し、`overlay` / `scale` / `pad` で合成した。

6. 成果物と再現性

種別	パス
テストケース定義	`deliverables/tests/cases.json`
実測値	`deliverables/tests/results/pub400.json`, `java.json`, `ipad.json`
比較結果	`deliverables/tests/results/compare.json`
フレーム	`deliverables/tests/results/pub400_frames.json`
ランナー	`run_pub400.py`, `run_java.js`, `run_ipad.sh`, `record_ipad.sh`
動画変換	`frames_to_video.js`, `overlay_ipad.js`